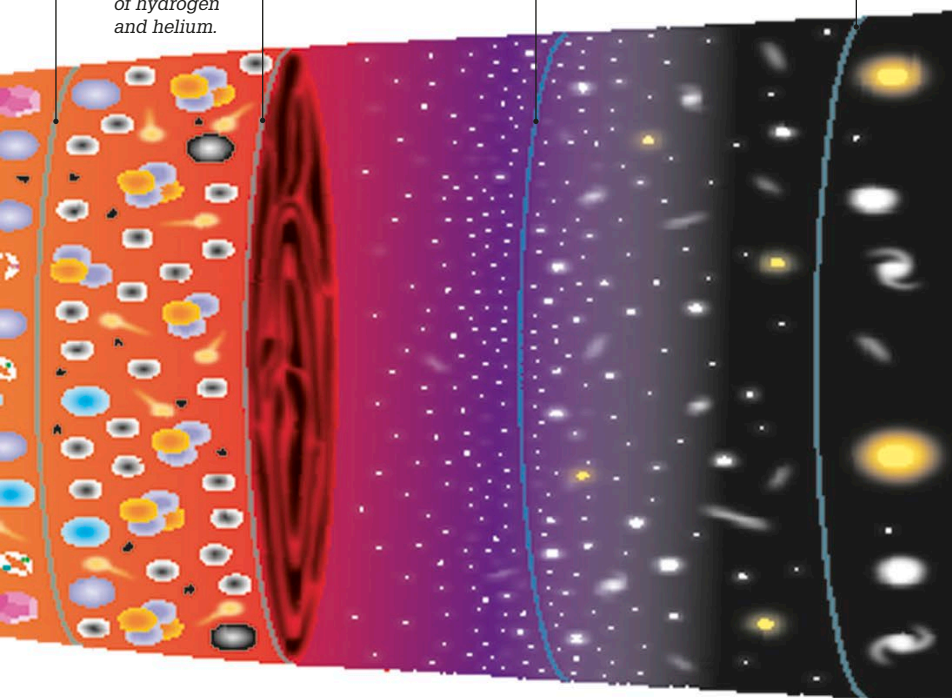


4. 3 minutes:
The first
protons and
neutrons form
and create
atomic nuclei
of hydrogen
and helium.

5. 380,000 years:
The Universe has cooled
enough for the first atoms
to form. Space becomes
transparent and light
can shine.

6. 1 billion years:
The earliest stars and
galaxies (systems of
stars) have formed.

7. Present day:
The Universe is
thought to be
approximately
13.8 billion
years old.



The Big Bang theory proposes that the Universe expanded from an individual point, called a singularity.

1934

Catseye

English roadworker Percy Shaw patented a road safety device after noticing how a cat's eyes reflected light. Still used today, the "Catseye" consists of lenses set in a rubber and metal dome that is fitted into the road. The lenses reflect light from a vehicle's headlights to illuminate the middle of the road and sometimes the boundaries between road lanes, without using any power.



1935



1933

Frequency Modulation radio

American engineer Edwin Howard Armstrong invented the first practical FM (Frequency Modulation) radio. Compared to AM (Amplitude Modulation) radios of the time, FM offered a clearer, higher-quality signal with less noise and interference from nearby electrical equipment or storms. The first FM radio stations began broadcasting in the US in the late 1930s.

Tuning dials enable
user to switch to the
correct frequency to
receive a radio signal.

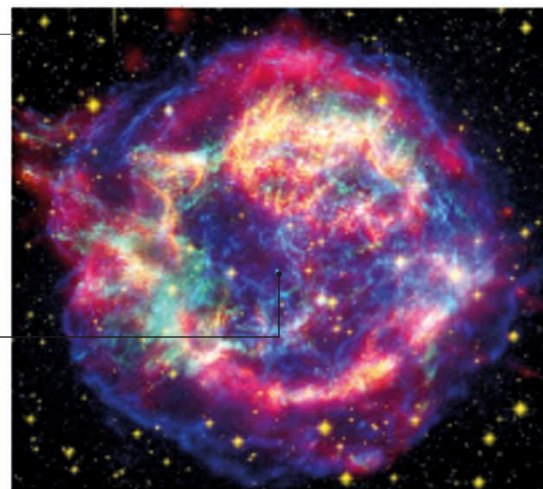
The case contains electric
circuits and six vacuum
tubes that help to make
the radio signal louder.



Armstrong's suitcase
FM radio receiver

The speaker
horn
broadcasts
the signal
as sound.

Neutron star glows
turquoise in this
composite of images
taken at different
wavelengths.



Remains of a supernova
containing a neutron star

1934

Exploded stars

Swiss astronomer Fritz Zwicky and German astronomer Walter Baade proposed that neutron stars (dense, collapsed stars mainly made of neutrons) form from the remains of colossal star explosions, which they called supernovae. A supernova occurs when a giant star exhausts its hydrogen fuel, collapses in on itself, and then rebounds violently to explode. Zwicky went on to discover the remains of 120 supernovae.